COMBINATIONS: We define symbol ${}^nC_r$ to mean number of ways of choosing r number of objects out of a total of n number of objects. You will learn a great deal about such problem under a very interesting branch of mathematics called combinatorics during your study of JEE mathematics. However here we shall state the important result without getting in to the proof. The result is as follows:

$$\boxed{{}^nC_r = \frac{n!}{r!(n-r)!}}$$

Here $n!$ is read as factorial n it means product of all natural numbers from 1 to n.
For instance $4! = 4 \times 3 \times 2 \times 1 = 24$. We stated the meaning of notation ${}^nC_r$ because it will be required for stating binomial theorem.

BINOMIAL THEOREM (NATURAL NUMBER EXPONENTS):

$$(1+x)^n = 1 + {}^nC_1 x^1 + {}^nC_2 x^2 + {}^nC_3 x^3 + \cdots {}^nC_n x^n$$

$$\Rightarrow \boxed{(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!} + \cdots x^n}$$

$(a+b)^n = a^n (1+x)^n$ where $x = b/a$ now using above expansion, we will get:

$$(a+b)^n = a^n \left[1 + n\frac{b}{a} + \frac{n(n-1)}{2!}\left(\frac{b}{a}\right)^2 + \cdots \left(\frac{b}{a}\right)^n\right]$$

$$\Rightarrow \boxed{(a+b)^n = a^n + na^{n-1}b^1 + \frac{n(n-1)}{2!}a^{n-2}b^2 + \cdots b^n}$$

EXAMPLE: Evaluate $(1+x)^4$ using binomial expansion.

Solution: $(1+x)^4 = 1 + 4x + \frac{4\times3}{2!}x^2 + \frac{4\times3\times2}{3!}x^3 + \frac{4\times3\times2\times1}{4!}x^4$

$= 1 + 4x + 6x^2 + 4x^3 + x^4$ Ans.

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Binomial Theorem (Negative and Fractional Exponents):

$$(1+x)^n = 1 + nx + \frac{n(n-1)x^2}{2!} + \frac{n(n-1)(n-2)}{3!} + \dots \text{ Infinite terms.}$$

→ Note that this is also true for natural number exponents, however with natural numbers, we will come across $(n-n)$ product after a certain term, thus all terms beginning with this go to zero and we are left with a finite series.

Example: Find $(1+x)^{-1}$ up to third order approximation (i.e x^3)

Solution: $(1+x)^{-1} = 1 + (-1)x + \frac{(-1)(-2)x^2}{2!} + \frac{(-1)(-2)(-3)}{3!}x^3 + \dots$

i.e $(1+x)^{-1} = 1 - x + \frac{x^2}{2} - x^3 + \dots$ - Higher order terms.

Binomial Approximation: In the expansion of $(1+x)^n$ if $x << 1$, we know that the term containing x^2 will be significantly smaller than the term containing x^1 and x^3 term will be still smaller, thus to get an approximate value of expansion we can drop higher order terms. Thus:

$$(1+x)^n \approx (1+nx)$$ ← First order approximation, Similarly:

$$(1+x)^n \approx 1 + nx + \frac{n(n-1)}{2}x^2$$ ← Second order approximation.

Example: Evaluate $\sqrt{1.1}$ (a) using first order approximation (b) Using second order approximation (c) Using Calculator (to four decimals).

Solution: (a) $(1+0.1)^{1/2} \approx 1 + 0.1/2 = 1.05$ (First order)

(b) $(1+0.1)^{1/2} \approx 1 + \frac{0.1}{2} + \frac{0.5(-0.5)}{2}(0.1)^2 = 1.049$

(c) $\sqrt{1.1} = 1.0488$

Thus we see that even the first order approximation is fairly accurate even though 0.1 is 10% of 1. Depending on our requirement

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we can increase the number of terms in expansion to get progressively more and more accurate.

EULER'S NUMBER (e): Just like π, e is another constant which has a great significance in physics and natural phenomena, its value up to three decimals is given as $e \approx 2.718$, though just like π it is a never ending and non recurring decimal number (Irrational number). It can be approximated as 19/7 (Less than 0.15% error) or 68/25 (less than 0.07% error).

NATURAL LOGARITHM ($\ln(x)$): Logarithms to base e are called natural logarithms i.e.

$$\ln(x) = \log_e(x)$$

Often we might need to change logarithms from base 10 to base e. It is useful to memorize the following result:
$\ln(10) \approx 2.303$ also $\ln(2) = 0.693$. The last result is especially useful in study of radioactivity and chemical kinetics.

SOME OTHER USEFUL EXPANSIONS:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} + \dots$$

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!}$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

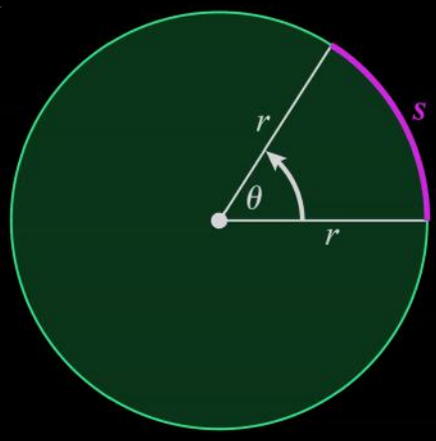
→ For very small angles i.e $x << 1$, $\sin(x) \approx x$ (x must be put in radians. For conversion $180° = \pi$ radians).

→ For very small angles $\cos(x) \approx 1 - \frac{x^2}{2}$ (Up to second order).

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TRIGONOMETRY:

RADIAN MEASURE OF ANGLE: If s is the length of arc of a circle of radius r then we define its angle in radians as:



$$\theta = s/r$$

Now for complete circumference, $\theta = 2\pi r/r$ i.e 2π radians thus $\boxed{2\pi \text{ radian} = 360^\circ}$ thus for converting any angle from degrees to radians, we need to multiply by a factor of $2\pi/360$ i.e $\pi/180$ thus:

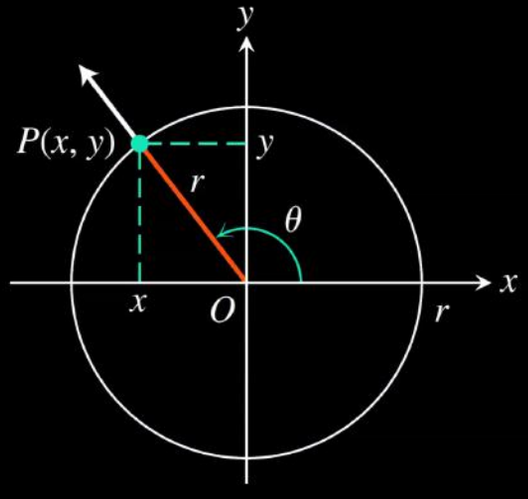
$$\theta_{rad} = \theta^\circ \times \frac{\pi \text{ rad}}{180^\circ}$$

In sciences and mathematics, default unit of angle is radians. Unless otherwise specified, whenever we say that certain angle measure is x we mean x radians. Thus if an angle is $\pi/2$ it means 90°. Following degree to radian conversions should be committed to memory for instant recall:

$$90^\circ \to \pi/2,\ 60^\circ \to \pi/3,\ 45^\circ \to \pi/4,\ 30^\circ \to \pi/6$$
$$120^\circ \to 2\pi/3,\ 135^\circ \to 3\pi/4,\ 150^\circ \to 5\pi/6,\ 180^\circ \to \pi$$

GENERALISED TRIGONOMETRIC RATIOS:

In junior classes we study trigonometric ratios in reference to a right triangle, which restricts our definition of trigonometric ratios to acute angles. For general angles we find trigonometric ratios are defined using following method.



① Plot the first side of angle along $+x$ axis

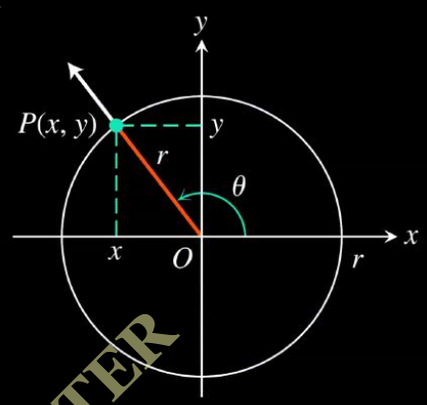
② Measure the angle (say θ) in anticlockwise sense from $+x$ direction and plot a

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point in this direction (say P) and join this with origin to form the other side of the angle.

③ Now let (x, y) be the coordinates of this point then we define trigonometric ratios for this angle θ as follows:

sine:	$\sin\theta = \frac{y}{r}$	**cosecant:**	$\csc\theta = \frac{r}{y}$
cosine:	$\cos\theta = \frac{x}{r}$	**secant:**	$\sec\theta = \frac{r}{x}$
tangent:	$\tan\theta = \frac{y}{x}$	**cotangent:**	$\cot\theta = \frac{x}{y}$



→ r is always considered as positive.

Signs Of Trigonometric Ratios In Four Quadrants:

Quadrant →	I	II	III	IV
$\sin\theta$	+	+	−	−
$\cos\theta$	+	−	−	+
$\tan\theta$	+	−	+	−
$\text{cosec}\,\theta$	+	+	−	−
$\sec\theta$	+	−	−	+
$\cot\theta$	+	−	+	−

→ Since $\sin(\theta)$ is y/r it is positive whenever y is +ve (I & II)

→ Since $\cos(\theta)$ is x/r it is positive whenever x is +ve (I & IV)

→ Since $\tan\theta$ is y/x it is positive whenever x & y have same sign (I & III).

Basic Trigonometric Identities:

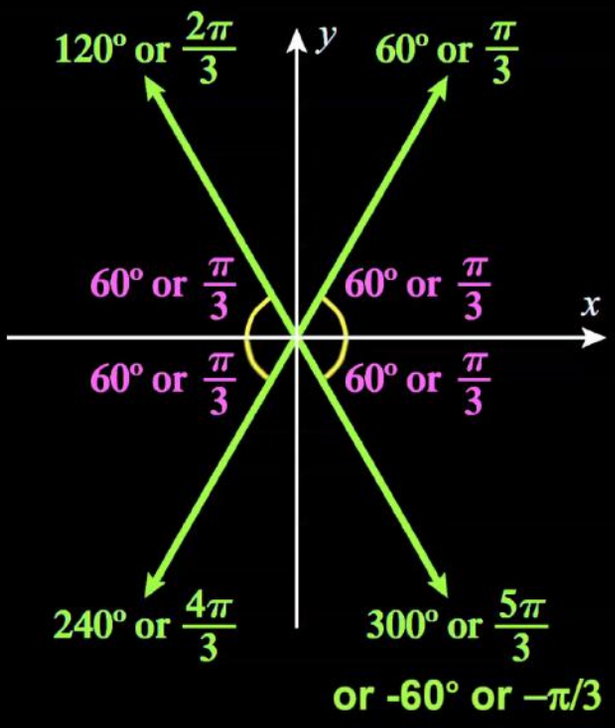
$\sin^2\theta + \cos^2\theta = 1$ $\quad$ $\sec^2\theta = 1 + \tan^2\theta$ $\quad$ $\text{cosec}^2\theta = 1 + \cot^2\theta$

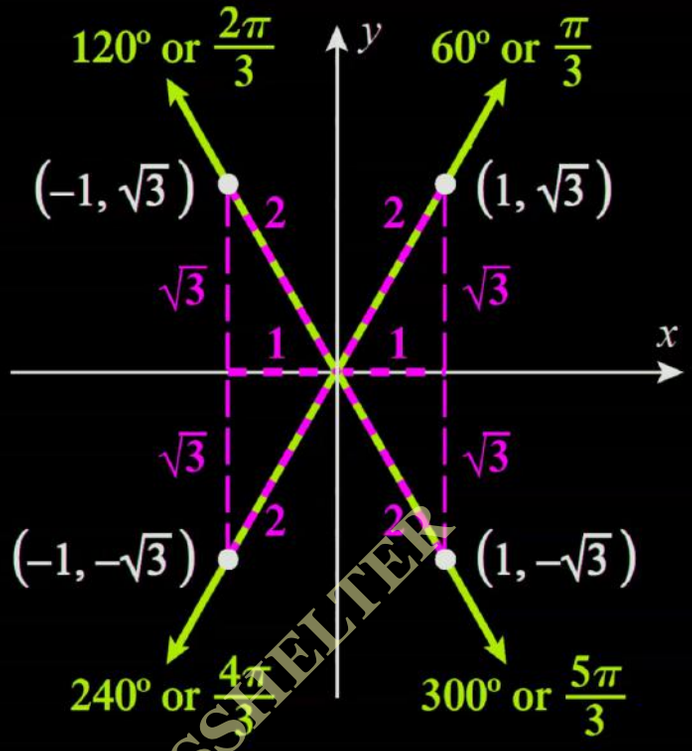
$\sin(A+B) = \sin A\cos B + \cos A\sin B$ $\quad$ $\cos(A+B) = \cos A\cos B - \sin A\sin B$

$\tan(A+B) = (\tan A + \tan B)/(1 - \tan A\tan B)$

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Quickly Evaluating Trigonometric Ratios In II, III and IV Quadrants:





Consider four angles in four quadrants, 60°, 120° (or 180°−60°), 240° (or 180°+60°) and −60° (or 300°). We can clearly see from the above figure that magnitudes of trigonometric ratios for all the four angles will be same however their signs will depend on the quadrant they are in. Thus:

$\sin(120°) = \sin(180°-60°) = \sin 60° = \sqrt{3}/2$ Quadrant II

$\sin(240°) = \sin(180°+60°) = -\sin 60° = -\sqrt{3}/2$ Quadrant III

$\sin(300°) = \sin(-60°)$ or $\sin(360°-60°) = -\sin 60° = -\sqrt{3}/2$ Quadrant IV

Using this method we can also find the other trigonometric ratios.

i.e $\cos(120°) = -\cos 60°$, $\cos 240° = -\cos 60°$, $\cos(-60°) = \cos 60°$ and

$\tan(120°) = -\tan 60°$, $\tan 240° = \tan 60°$ and $\tan(-60°) = -\tan 60°$.

This observations leads to following formulae for evaluating trigonometric ratios in various quadrants:

$\sin(180°-\theta) = \sin\theta$	$\cos(180°-\theta) = -\cos\theta$	$\tan(180°-\theta) = -\tan\theta$
$\sin(180°+\theta) = -\sin\theta$	$\cos(180°+\theta) = -\cos\theta$	$\tan(180°+\theta) = \tan\theta$
$\sin(-\theta) = -\sin\theta$	$\cos(-\theta) = \cos\theta$	$\tan(-\theta) = -\tan\theta$

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EXAMPLE: Find Sin, Cos and Tan for angle $\theta = 210°$.

Solution: We observe $210° = 180° + 30°$ thus:

$\sin(210°) = -\sin 30° = -1/2$ $\quad \cos(210°) = -\cos 30° = -\sqrt{3}/2$

and $\tan(210°) = \tan 30° = 1/\sqrt{3}$ Ans.

Think: What are the three trigonometric ratios for $\theta = 5\pi/6$ (i.e 150°)

Solution: $150° = 180° - 30°$ $\Rightarrow$ $\sin(150°) = \sin 30° = 1/2$, $\cos(150°) = -\cos 30° = -\sqrt{3}/2$ and $\tan 150° = -\tan 30° = -1/\sqrt{3}$ Ans.

EXAMPLE Find all values of θ where $0° < \theta < 360°$ for which

(a) $\sin\theta = -\sqrt{3}/2$ (b) $\cos\theta = -\sqrt{3}/2$.

Solution: (a) Since Sin is -Ve, we must have an angle in 3rd or 4th Quadrant, also $\sin 60° = \sqrt{3}/2$, thus our required angles are $\theta = 180° + 60°$ i.e 240° and $360° - 60° = 300°$ Ans.

(b) Since $\cos\theta < 0$ $\Rightarrow$ θ must be in II or III Quadrant. Also, $\cos 30° = \sqrt{3}/2$ thus our required angles are $180° + 30°$ i.e 210° and $180° - 30°$ i.e 150° Ans.

INVERSE TRIGONOMETRIC FUNCTIONS: When we write the expression $\sin^{-1}(x)$ we mean an angle whose Sine is x. For example $\sin^{-1}(1/2) = 30°$ if our range excludes second quadrant. However if the second quadrant were also included then $\sin^{-1}(1/2)$ would be ambiguous to mean either 30° or 150°.

→ Domain means allowed input values for a function (for instance $\sin^{-1}(x)$ has the domain $[-1, 1]$, where rectangular brackets mean that -1 and 1 are also included in domain. For instance if we have a function $1/(x-1)(x-2)$, its domain is: $(-\infty, 1) \cup (1, 2) \cup (2, \infty)$, these round brackets mean open intervals, i.e the end point values are excluded.

→ Range means the values a function may take for all possible input values in its domain. For range of $\sin(x)$ is $[-1, 1]$.

Think: What is domain of function $y = \sqrt{x}$?

Solution: For a real valued function, x must be non-negative thus Domain is $x \in [0, \infty)$ and Range is $[0, \infty)$

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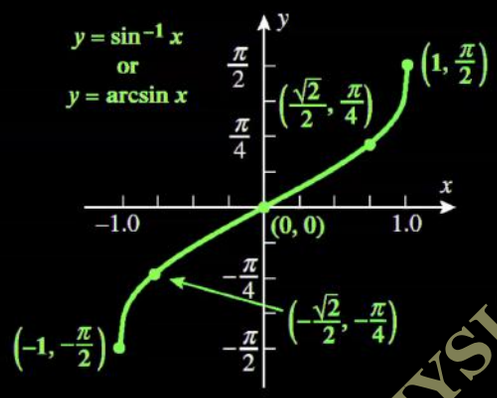
Range Of Inverse Trigonometric Functions:

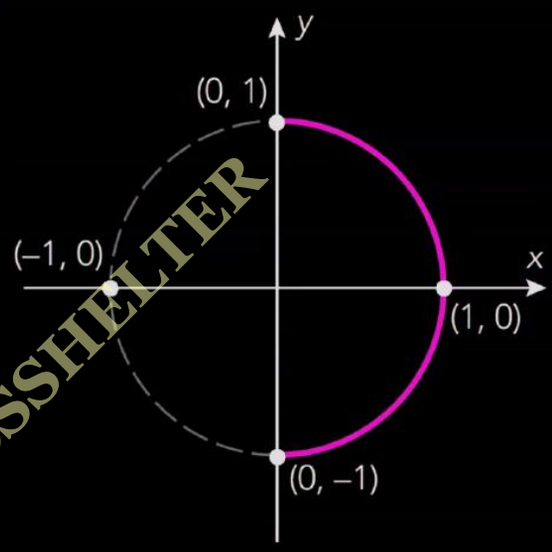
As we discussed in previous section that without restricting the ranges, inverse trigonometric functions would have ambiguous meanings. To avoid this difficulty we restrict their ranges as per following scheme.

$Sin^{-1}(x)$:

$y = \sin^{-1}x$ Domain: $[-1, 1]$ Range: $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

x	y
-1	$-\frac{\pi}{2}$
$-\frac{\sqrt{2}}{2}$	$-\frac{\pi}{4}$
0	0
$\frac{\sqrt{2}}{2}$	$\frac{\pi}{4}$
1	$\frac{\pi}{2}$



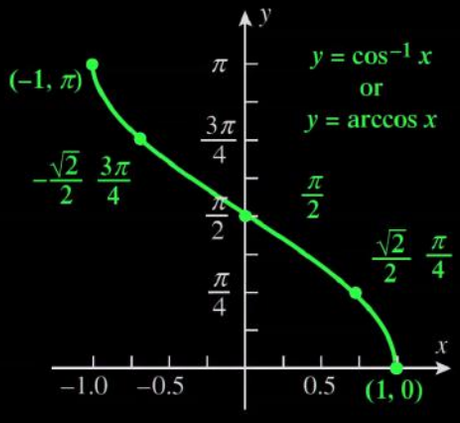


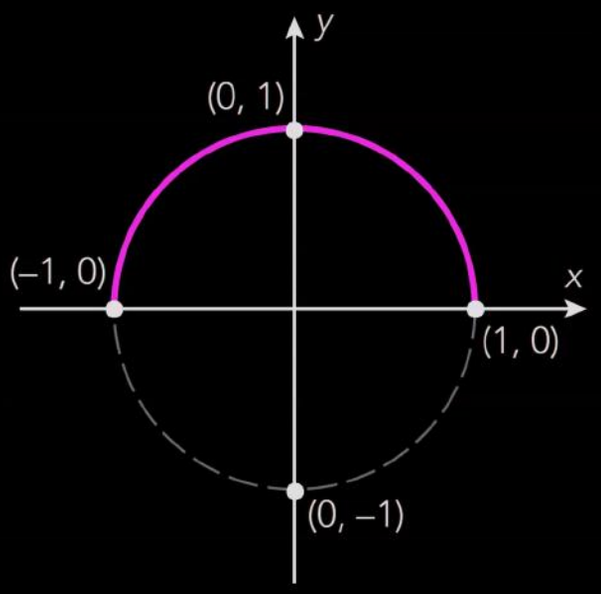
→ Whenever we write $Sin^{-1}(x)$ we always mean angle lying between $-\pi/2$ and $+\pi/2$ i.e between fourth quadrant and first quadrant. (For negative values it will be a fourth quadrant angle).
For instance $Sin^{-1}(-1/2) = -\pi/6$ (or $-30°$).

$Cos^{-1}(x)$

$y = \cos^{-1}x$ Domain: $[-1, 1]$ Range: $[0, \pi]$

x	y
-1	π
$-\frac{\sqrt{2}}{2}$	$\frac{3\pi}{4}$
0	$\frac{\pi}{2}$
$\frac{\sqrt{2}}{2}$	$\frac{\pi}{4}$
1	0





→ Whenever we write $Cos^{-1}(x)$ we always mean an angle between $0°$ and $180°$ (for negative x, between $90°$ and $180°$).

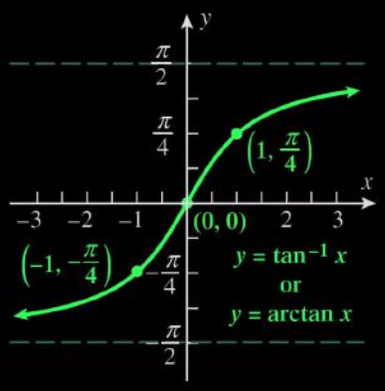
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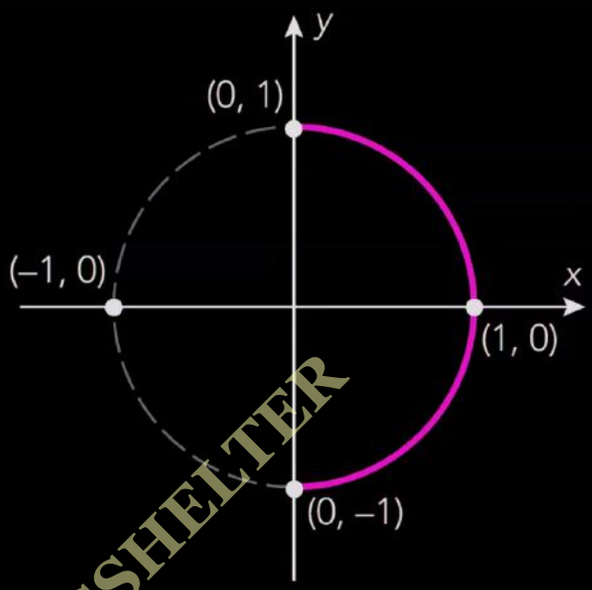
→ For instance $\cos^{-1}(-\sqrt{3}/2) = 150°$. (For positive value of x, we mean an angle between 0° and 90°)

$\tan^{-1}(x)$:

$y = \tan^{-1}x$ Domain: $(-\infty, \infty)$ Range: $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

x	y
$-\infty$	$-\frac{\pi}{2}$
-1	$-\frac{\pi}{4}$
0	0
1	$\frac{\pi}{4}$
∞	$\frac{\pi}{2}$





→ Whenever we say $\tan^{-1}(u)$ we always mean an angle between $-\pi/2$ and $\pi/2$ (or $-90°$ and $+90°$).

→ For negative x, $\tan^{-1}(x)$ gives a value in fourth quadrant for instance $\tan^{-1}(-\sqrt{3}) = -\pi/3$ or $-60°$.

→ For Cosec^{-1}, Sec^{-1} and Cot^{-1} functions, the principle ranges are same as those for $\sin^{-1}$, $\cos^{-1}$ and $\tan^{-1}$ respectively.

EXAMPLE: Express $\cos[\tan^{-1}(u)]$ in terms of u only.

Solution: $\tan^{-1}(u)$ is an angle in I or II quadrant thus its cosine must be a positive number. Now if $\theta = \tan^{-1}(u)$

$\Rightarrow \tan\theta = u \Rightarrow \sec^2\theta = 1+u^2 \Rightarrow \cos^2\theta = 1/(1+u^2)$

$\Rightarrow \cos\theta = 1/\sqrt{1+u^2}$ Ans.

Think: Write $\sin[\tan^{-1}(u)]$ in terms of u alone.

Solution: $\tan^{-1}(u)$ is an angle in either IV or I Quadrant, thus sign of its sine will be same as that of u. Now if

$\theta = \tan^{-1}u \Rightarrow u = \tan\theta = p/b$ (Now think $p = u$, $b = 1$)

$\Rightarrow h = \sqrt{1+u^2} \Rightarrow \sin\theta = \dfrac{u}{\sqrt{1+u^2}}$ Ans. [p → perpendicular, b → base, h → hypotenuse]

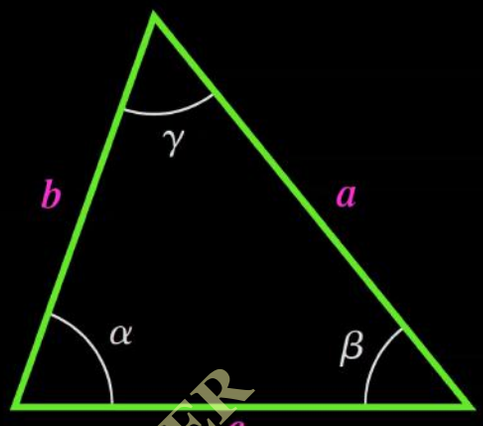
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Think: Find the value of (a) $\tan^{-1}(-1/\sqrt{3})$ (b) $\operatorname{cosec}^{-1}(-2/\sqrt{3})$ (c) $\cot^{-1}(-1/\sqrt{3})$

Ans: (a) $-30°$ (b) $-60°$ (c) $-60°$

SINE LAW: In any Arbitrary triangle:

$$\frac{a}{\sin\alpha} = \frac{b}{\sin\beta} = \frac{c}{\sin\gamma}$$



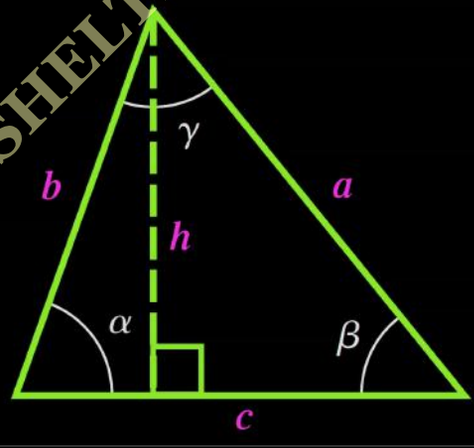
→ On simply stated any side divided by the sine of opposite angle is same.

PROOF OF SINE LAW:

Consider the triangle shown in adjacent figure. Here after drawing the altitude, we can clearly see that,

$h = b\sin\alpha = a\sin\beta$, thus:

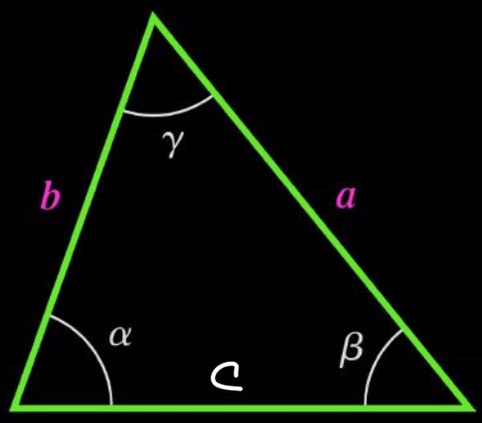
$\frac{a}{\sin\alpha} = \frac{b}{\sin\beta}$. Same way we can draw other altitudes to show it to be equal to $c/\sin\gamma$ also.



COSINE LAW: Consider the arbitrary triangle as shown in the figure. In any such triangle:

$$c^2 = a^2 + b^2 - 2ab\cos\gamma$$

Similarly, $a^2 = b^2 + c^2 - 2bc\cos\alpha$ and $b^2 = a^2 + c^2 - 2ac\cos\beta$



PROOF OF COSINE LAW: Consider the adjacent triangle:

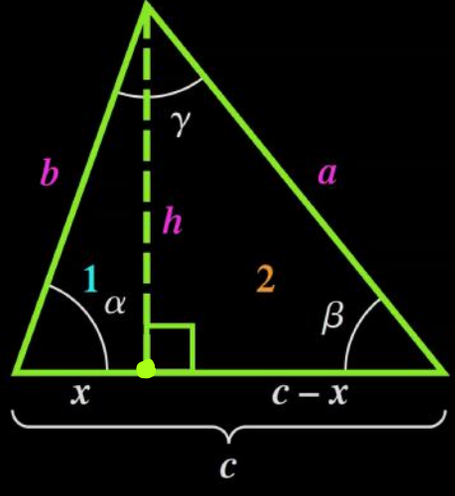
$$x^2 = b^2 - h^2 \quad (1)$$

$$(c-x)^2 = a^2 - h^2 \quad (2)$$

Now (2) − (1) yields: $c^2 - 2cx = a^2 - b^2$

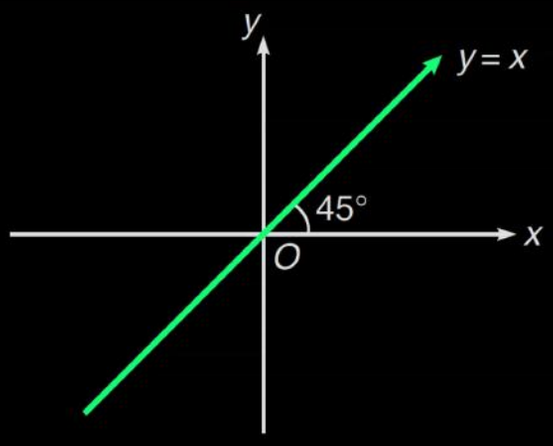
$\Rightarrow$ $a^2 = b^2 + c^2 - 2cx = b^2 + c^2 - 2bc\cos\alpha$

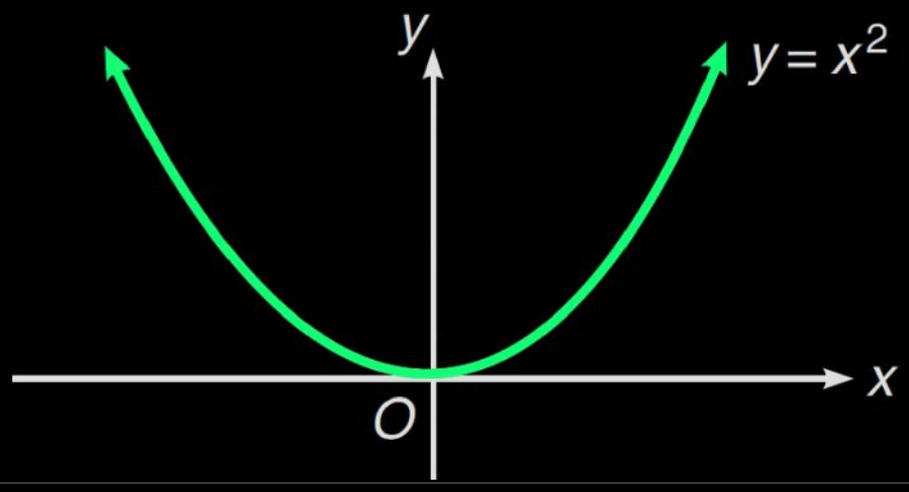
QED.



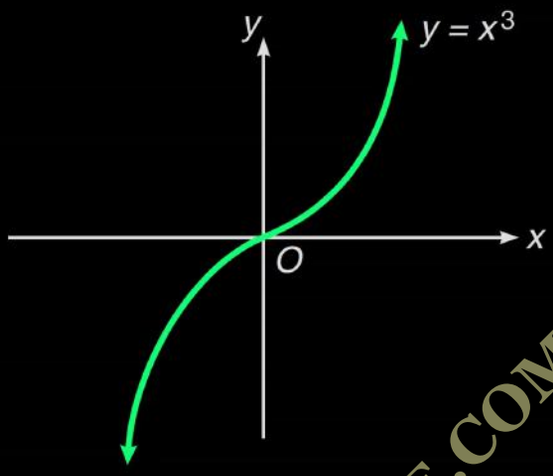
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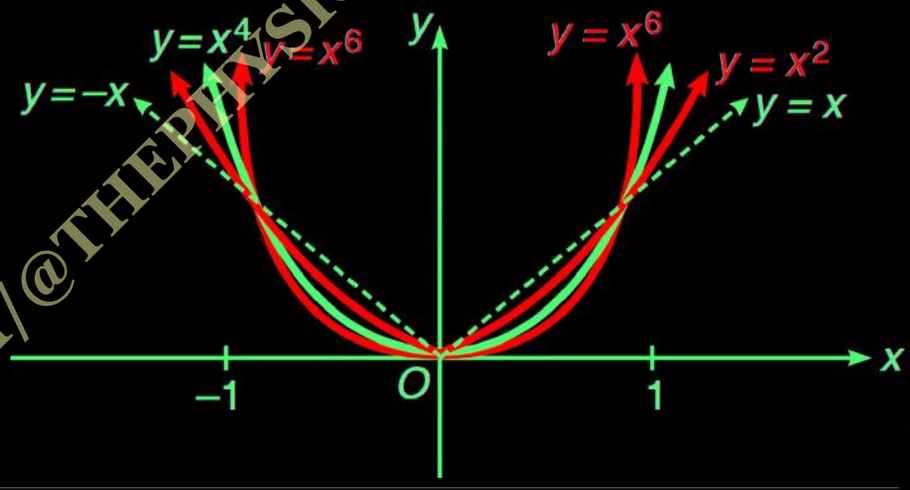
Some Simple Functions And Their Plots:





→ The function $y = f(x) = x^2$ is an even function which mean $f(-x) = f(x)$ for example $(-2)^2 = 2^2$. Whenever this happens the graph is symmetric about y axis (Left half is mirror image of right half).

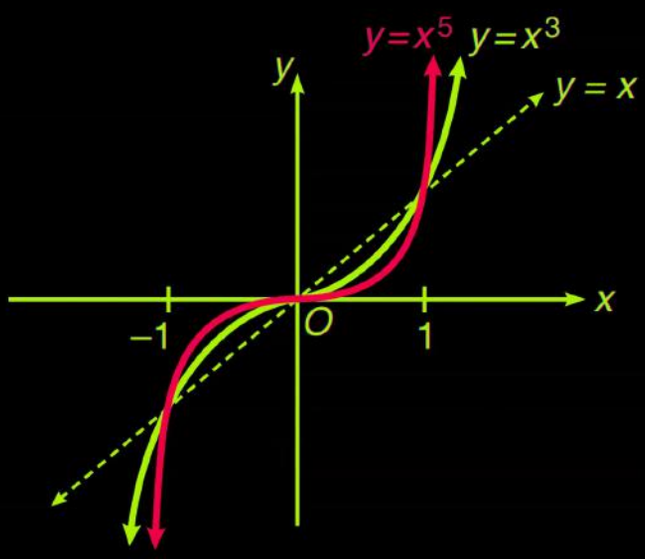


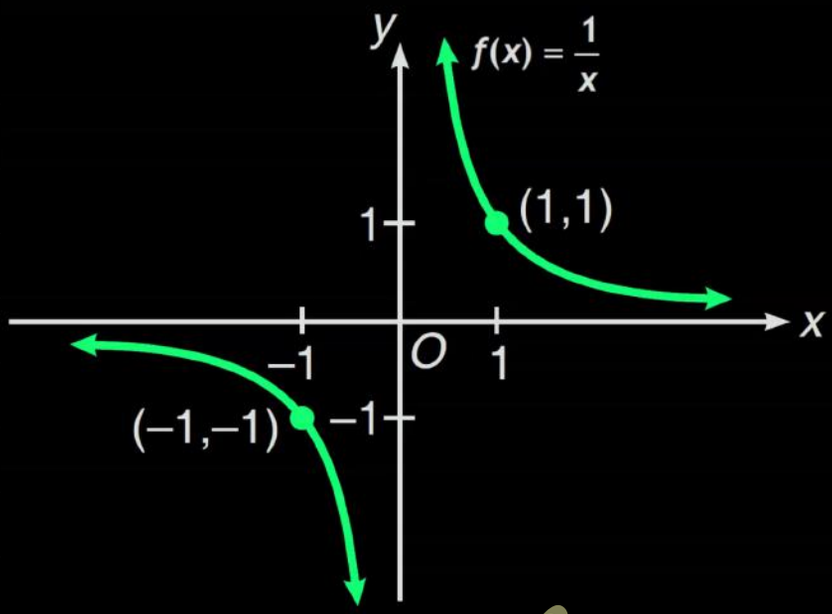


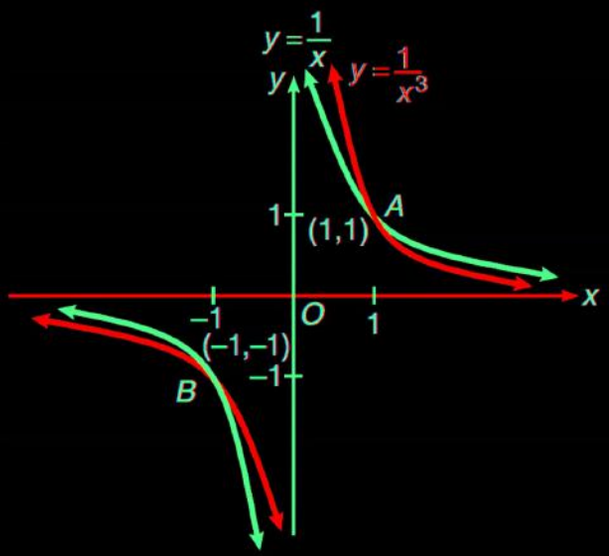
→ The function $y = f(x) = x^3$ is an odd function which means $f(-x) = -f(x)$ for example $(-3)^3 = -3^3$. Whenever this happens, this graph is symmetric about origin. This means, left part is mirror image of water image of right part.

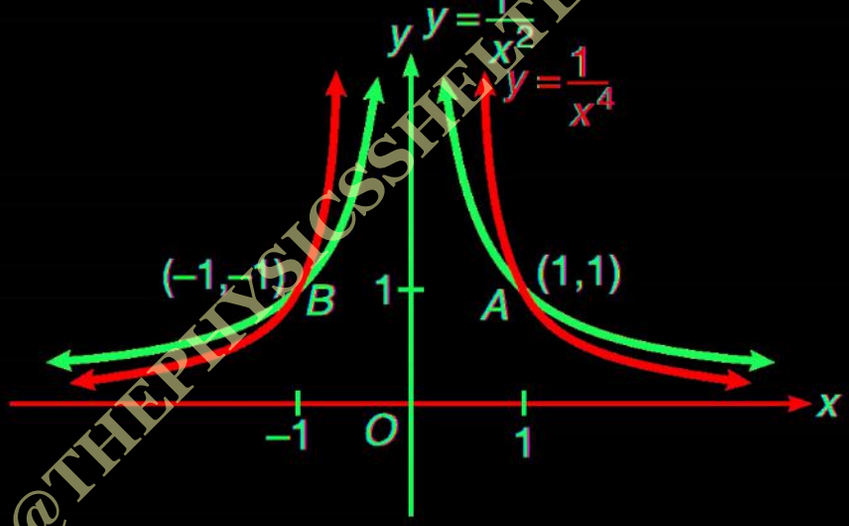
→ For functions of the form $y = f(x) = x^{2n}$, the value of y increases with x if $|x| > 1$ and decreases with x if $|x| < 1$.

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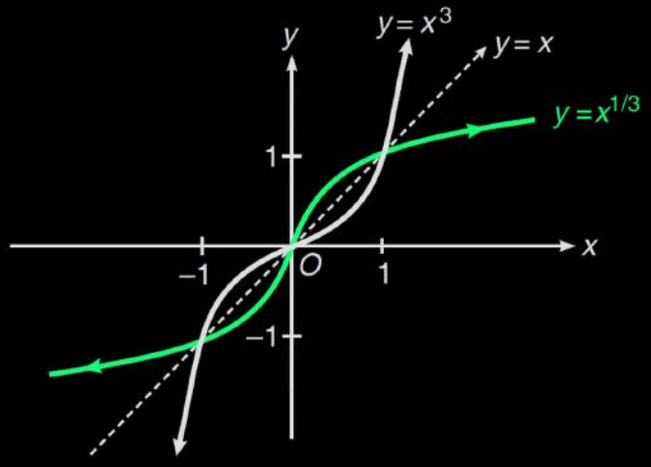


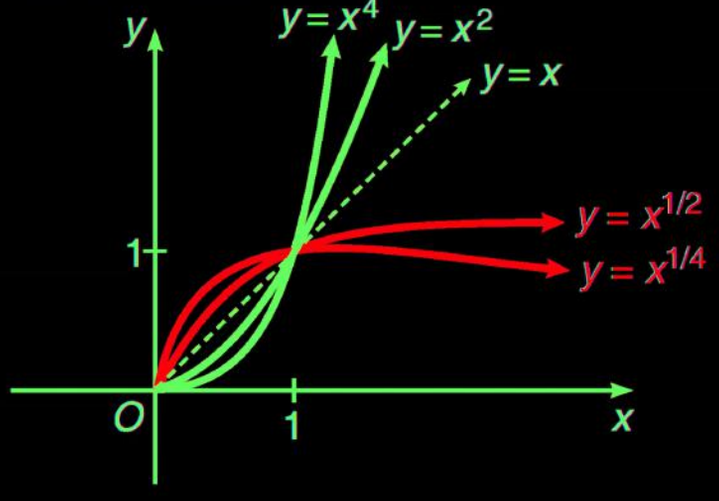




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Inverse Functions: A function obtained by swapping x and y is called the inverse function of the original function. For instance if we wish to find inverse function of $y=x^3$, we swap x & y to obtain $x=y^3$ or $y=x^{1/3}$ now this is our required inverse function. Inverse functions are mirror images of each other about line $y=x$ (i.e 45° line through I & III Quadrants).

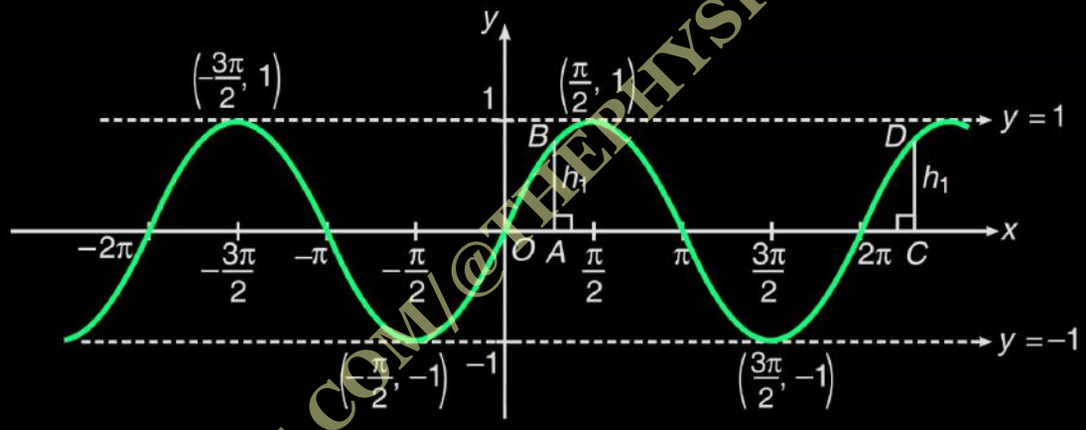




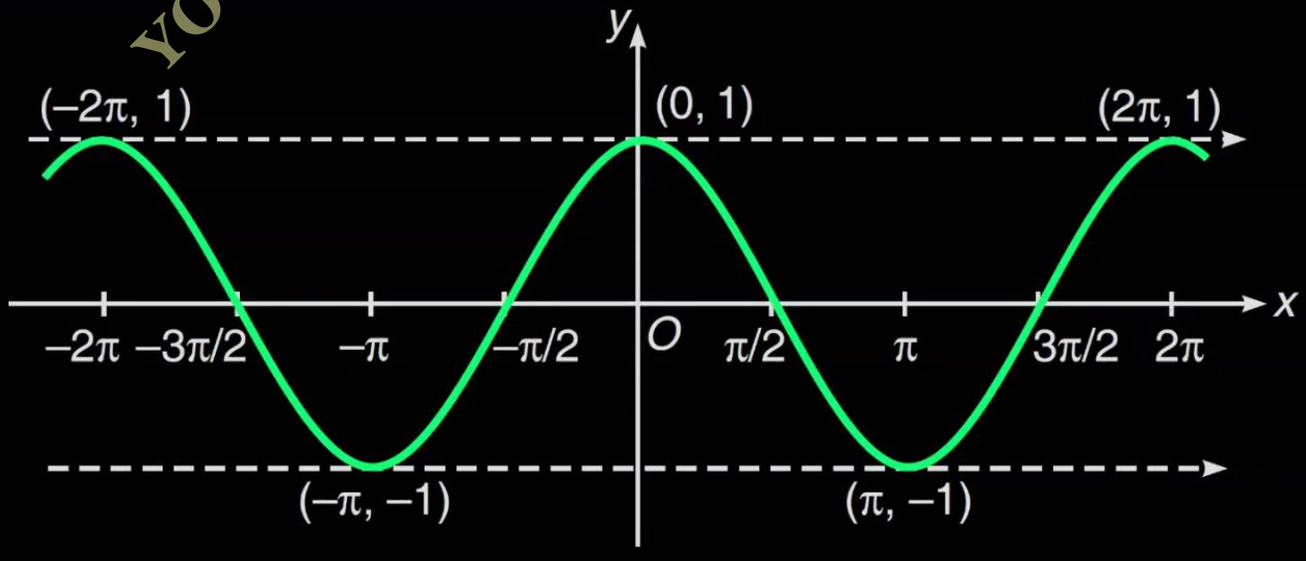
Periodicity Of A Function:

If $f(x) = f(x+T)$ for any arbitrary x and some particular minimum value of T, then we say that T is periodicity of the function. Now recall that all angles are measured from $+x$ axis and if any line is rotated by an angle 2π it returns to the same configuration in the x-y plane. Thus we can say that $\sin(x+2\pi) = \sin(x)$ for every value of x. Thus 2π is the period of the function $\sin(x)$ which means at all values of x separated by a distance 2π, this function will have exactly same value. Similarly $\cos(x)$, $\sec(x)$ and $\operatorname{cosec}(x)$ also have a time period of 2π. [Tan x & Cot x → π]. Also adding any integer multiple of time period should not make any difference to the value of the function, thus $\sin(x+2n\pi) = \sin(x)$.

Plot Of Sin(x):

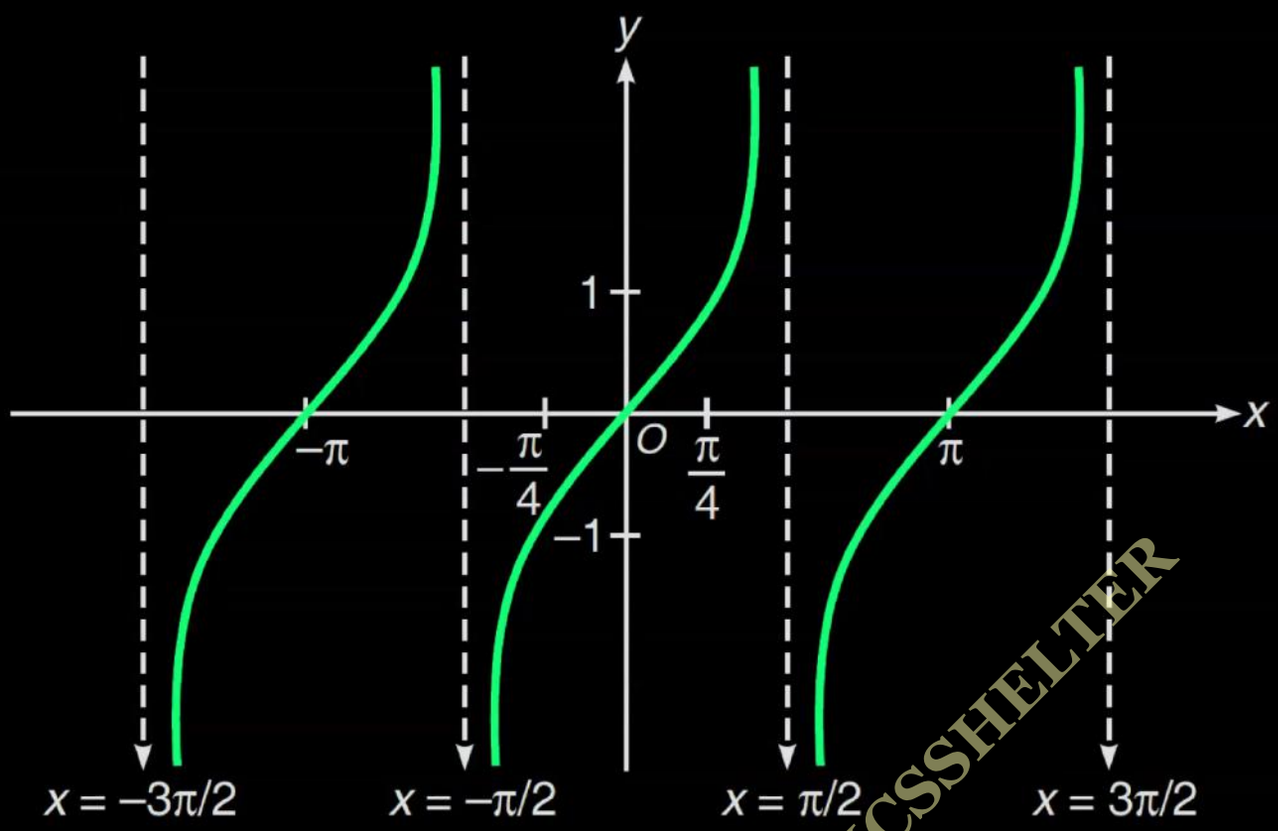


Plot Of Cos(x):

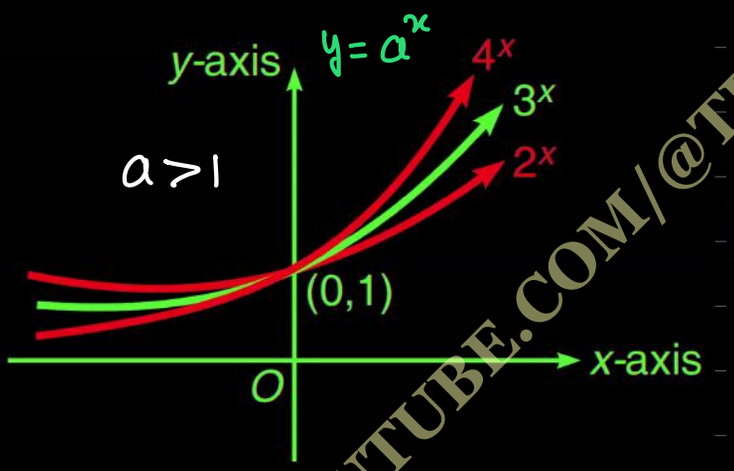


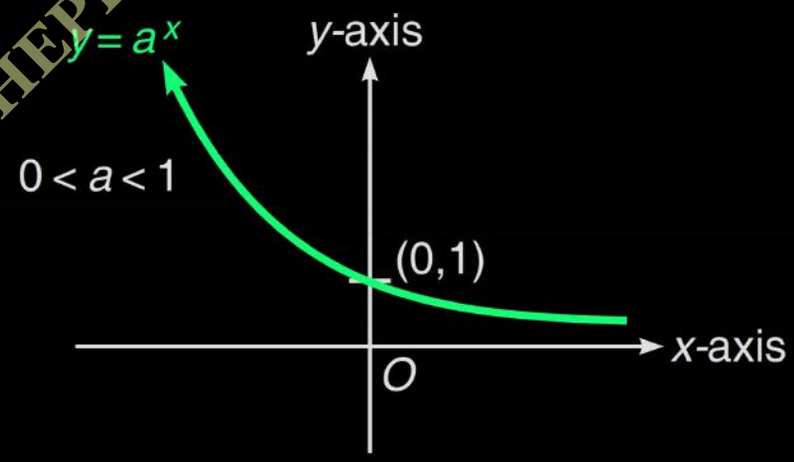
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PLOT OF Tan(x):

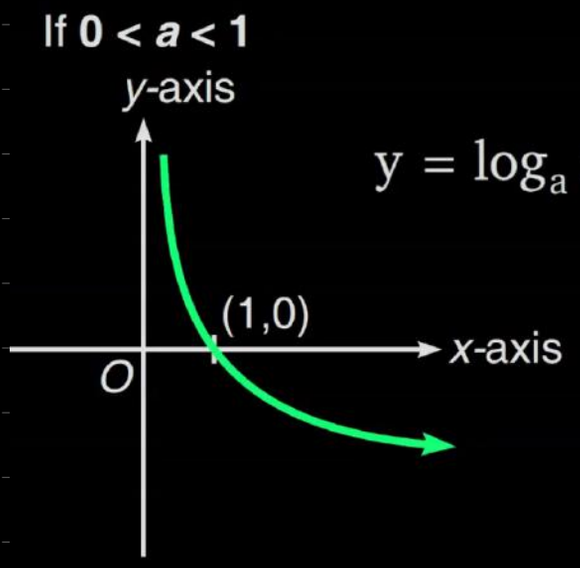


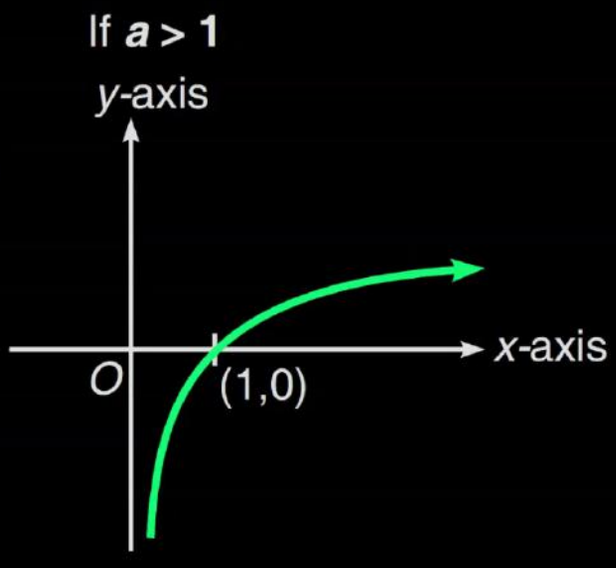
PLOTS OF EXPONENTIAL FUNCTION:





PLOTS OF LOGARITHM FUNCTION: (INVERSE EXPONENTIAL):





Exp: $y = a^x$

Swap:

$x = a^y$

$\Rightarrow \log_a x = y.$

Thus it is inverse of exponential function.

AMBARISH SIR

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TRANSFORMATIONS OF GRAPHS: Consider a function $y = f(x)$ and its modified version $y_2 = a\,f[b(x+c)] + d$. Then y_2 can be obtained by suitably modifying the shape of $f(x)$. The effect of parameters a, b, c and d are as follows:

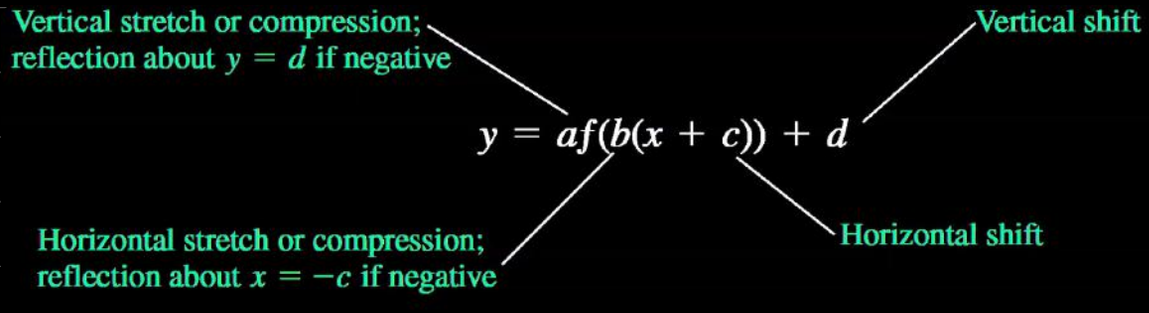
a: It represents vertical scaling factor. For instance if $y_1 = \sin(x)$ and $y_2 = 2\sin(x)$ then y_2 is twice as tall everywhere as y_1.

b: It represents horizontal scaling of $f(x)$. Thus if $y_1 = \sin(x)$ and $y_2 = \sin(2x)$, then y_2 will be a shrunk version of y_1 by a factor of 2. For whatever value $\sin(x)$ assumes for $x = \pi/2$, $\sin(2x)$ will assume same value for $x = \pi/4$. Thus for a general b, our graph will shrink by a factor of b.

c: It represents horizontal shifting of graph. For instance if $y_1 = \sin(x)$ and $y_2 = \sin(x + \pi/4)$ whatever value y_1 assumes for $x = \pi$, y_2 will assume same value for $x = \pi - \pi/4$. Thus our graph will be shifted $\pi/4$ units to the left. For a general c, the graph will shift by c units to the left).

d: It represents vertical shifting of graph. For example if $y_1 = \sin(x)$ and $y_2 = \sin(x) + 2$ then y_2 will be everywhere upward shifted from y_1 by 2 units. For a general d, this shift will be d units upwards.

→ All the four effects are summarized in the following diagram:



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Vertical and Horizontal Scaling and Reflecting Formulas

For $c > 1$, the graph is scaled:

$y = cf(x)$	Stretches the graph of f vertically by a factor of c.
$y = \frac{1}{c}f(x)$	Compresses the graph of f vertically by a factor of c.
$y = f(cx)$	Compresses the graph of f horizontally by a factor of c.
$y = f(x/c)$	Stretches the graph of f horizontally by a factor of c.

For $c = -1$, the graph is reflected:

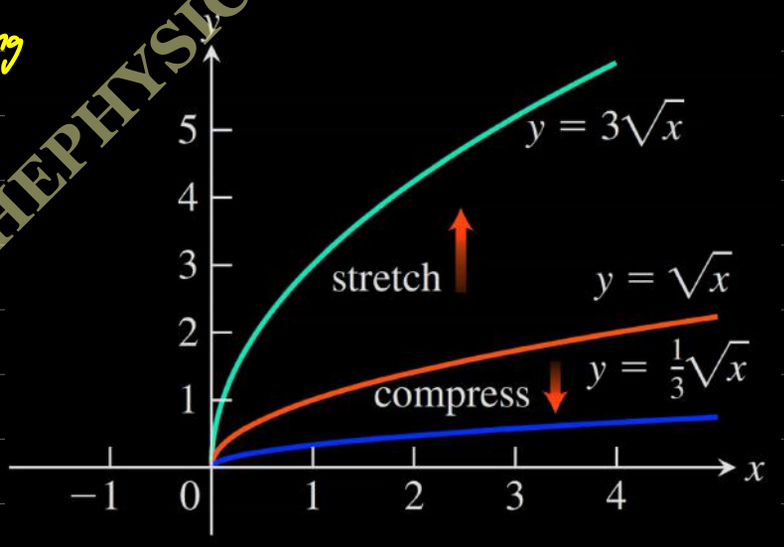
$y = -f(x)$	Reflects the graph of f across the x-axis.
$y = f(-x)$	Reflects the graph of f across the y-axis.

EXAMPLE: (THOMAS CALCULUS): In this example we shall study Horizontal and Vertical scaling and reflection of the graph $y = \sqrt{x}$.

(a) Vertical stretching and compressing by a factor of 3.

$y = 3\sqrt{x}$ stretches the $y = \sqrt{x}$ graph by a factor of 3.

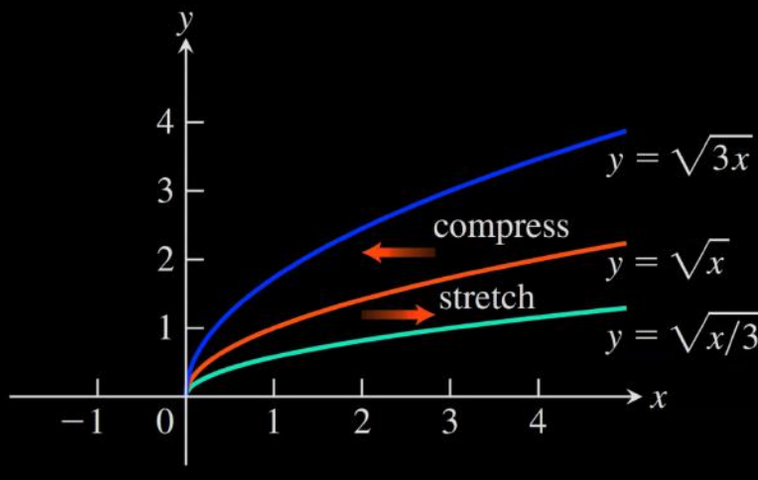
$y = \frac{1}{3}\sqrt{x}$ compresses the graph vertically by a factor of 3.



(b) Horizontally stretching and scaling by a factor of 3:

$y = \sqrt{3x}$ compresses the graph of $y = \sqrt{x}$ by a factor of 3

$y = \sqrt{\frac{1}{3}x}$ stretches the graph of $y = \sqrt{x}$ by a factor of 3 horizontally.

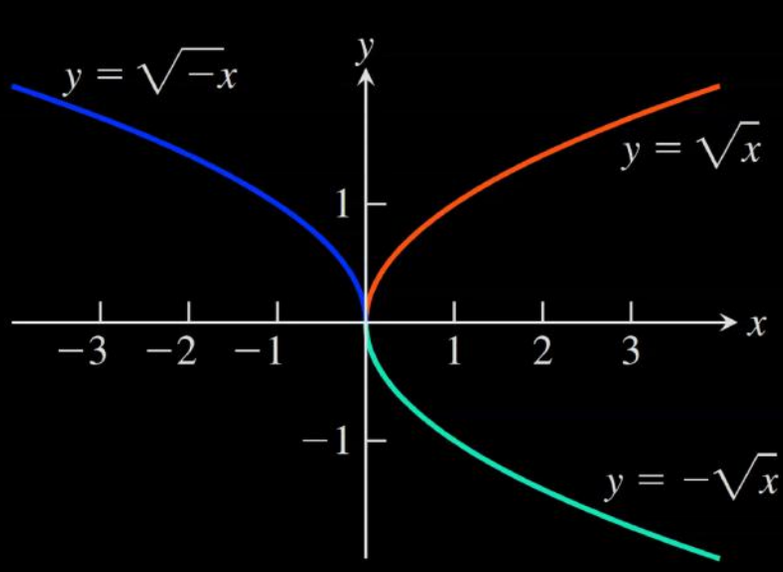


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(c) Reflections of graph $y = \sqrt{x}$ about x and y axes:

→ Function $y = -\sqrt{x}$ is reflection of $y = \sqrt{x}$ about x axis

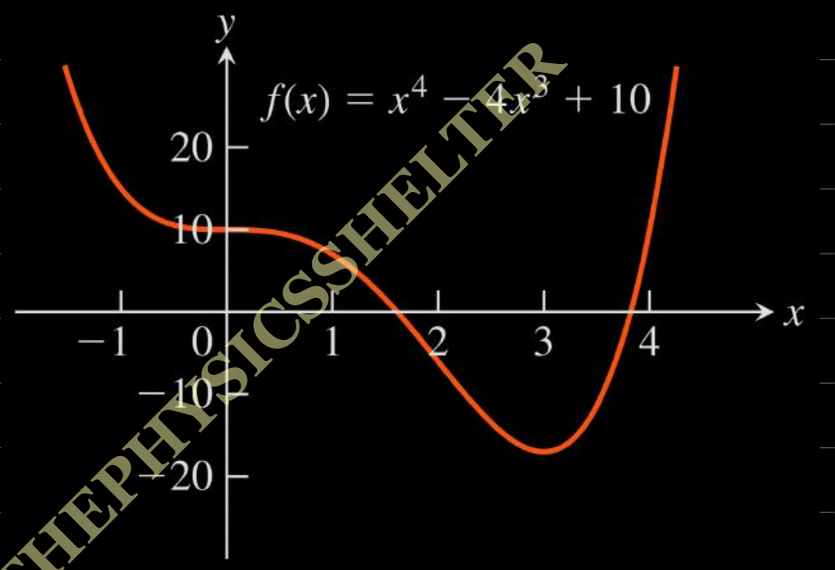
→ Function $y = \sqrt{-x}$ is reflection of $y = \sqrt{x}$ about y axis.



EXAMPLE: Given a function $y = f(x) = x^4 - 4x^3 + 10$ (See adjacent figure), find formulas to:

(a) Compress the graph horizontally by a factor of 2 followed by reflection about y axis

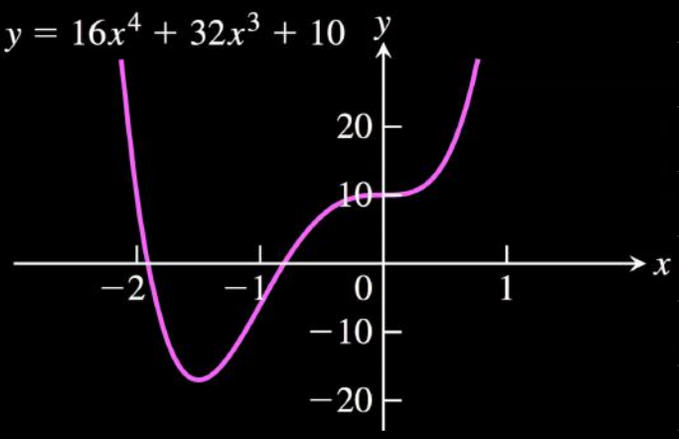
(b) Compress graph vertically by a factor of 2 followed by reflection about x axis.



Solution: (a) $f(2x)$ causes compression along x axis and then for reflection of about y axis we need $f(-2x)$. Now substituting $(-2x)$ for x in $f(x)$, we get required function:

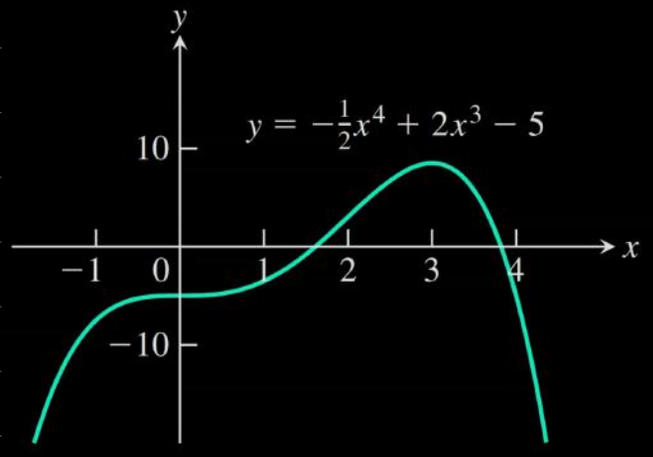
$y = g(x) = (-2x)^4 - 4(-2x)^3 + 10$

$= 16x^4 + 32x^3 + 10$ Ans.



(b) $\frac{1}{2} f(x)$ scales down the graph vertically by a factor of two and then, $-\frac{1}{2} f(x)$ reflects it about x axis thus, required function: $g(x) = -\frac{1}{2}[x^4 - 4x^3 + 10]$

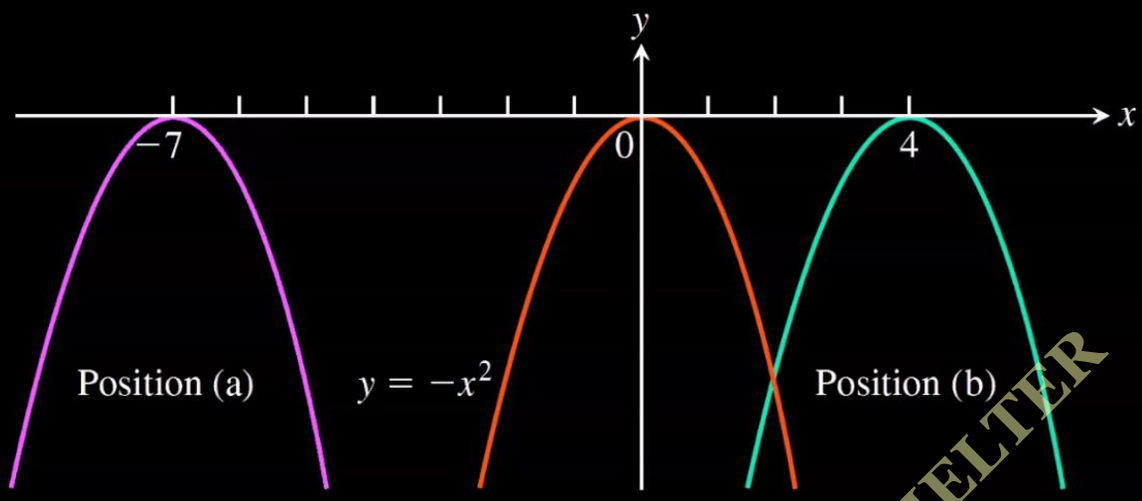
$\Rightarrow g(x) = -\frac{1}{2}x^4 + 2x^3 - 5$ Ans.



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Example: Thomas Calculus:

The accompanying figure shows the graph of $y = -x^2$ shifted to two new positions. Write equations for the new graphs.

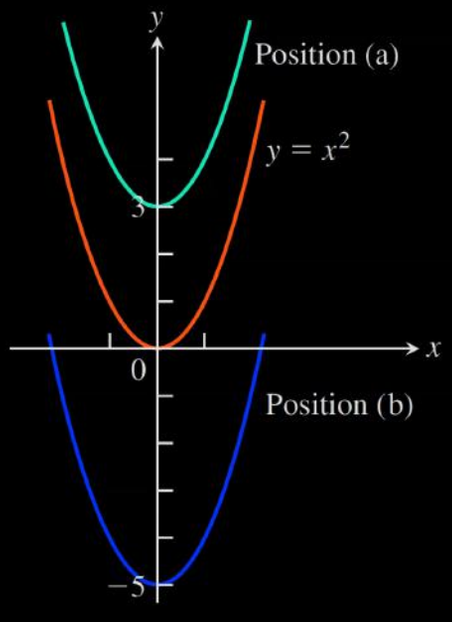


Solution: $f(x+a)$ represents a function shifted by a units to the left. Thus $f(x+7)$ will represent the position (a) and $f(x-4)$ represents the position (b). Thus required equation:

(a) $g(x) = -(x+7)^2$ (b) $h(x) = -(x-4)^2$.

Example: (Thomas Calculus):

The accompanying figure shows the graph of $y = x^2$ shifted to two new positions. Write equations for the new graphs.



Solution: $f(x)+a$ represents a function shifted upwards by a. Thus required equations:

Position (a): $y = x^2 + 3$

Position (b): $y = x^2 - 5$ Ans.

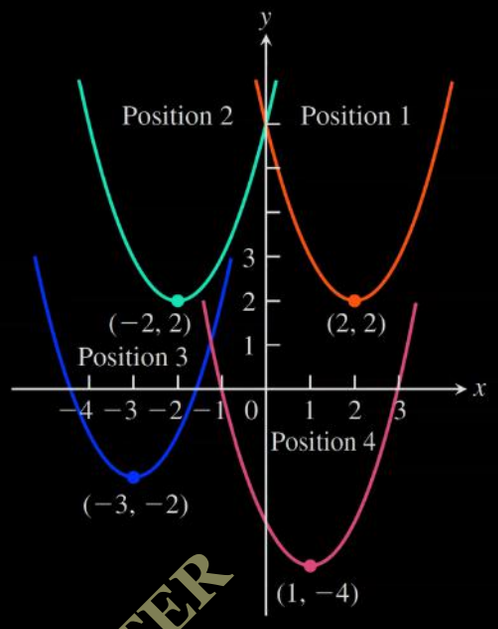
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EXAMPLE: (THOMAS CALCULUS):

Match the equations listed in parts (a)–(d) to the graphs in the accompanying figure.

a. $y = (x - 1)^2 - 4$ **b.** $y = (x - 2)^2 + 2$

c. $y = (x + 2)^2 + 2$ **d.** $y = (x + 3)^2 - 2$



Solution: All graphs are horizontal and vertical shiftings of $f(x) = x^2$. Thus position ① : $y = (x-2)^2 + 2$ i.e (b) (∵ Right and upward shift of 2 units each).

Similarly for ②: $y = (x+1)^2 + 2$ (c)

For ③, $y = (x+3)^2 - 2$ i.e (d) and ④: $y = (x-1)^2 - 4$ i.e (a) Ans.

PLOT OF ELLIPSE:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad (a^2 > b^2)$$

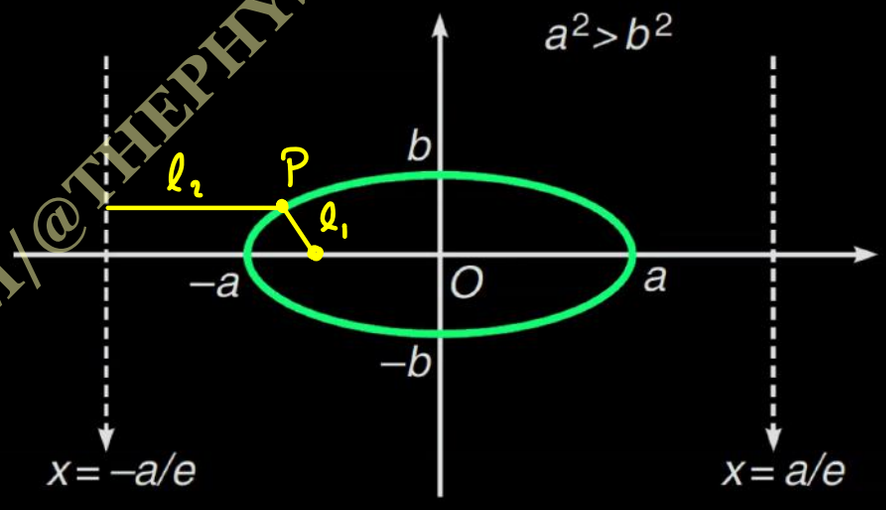
Centre : (0, 0)

Focus : (±ae, 0)

Vertex : (±a, 0)

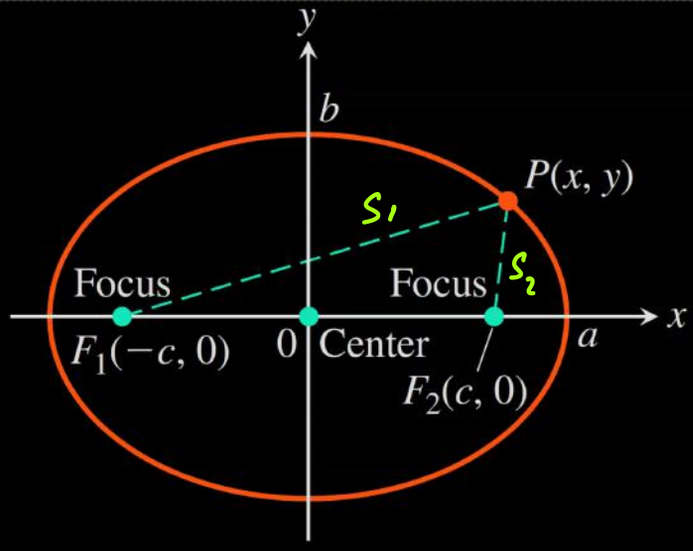
Eccentricity : $e = \sqrt{1 - \frac{b^2}{a^2}}$

Directrix : $x = \pm \frac{a}{e}$



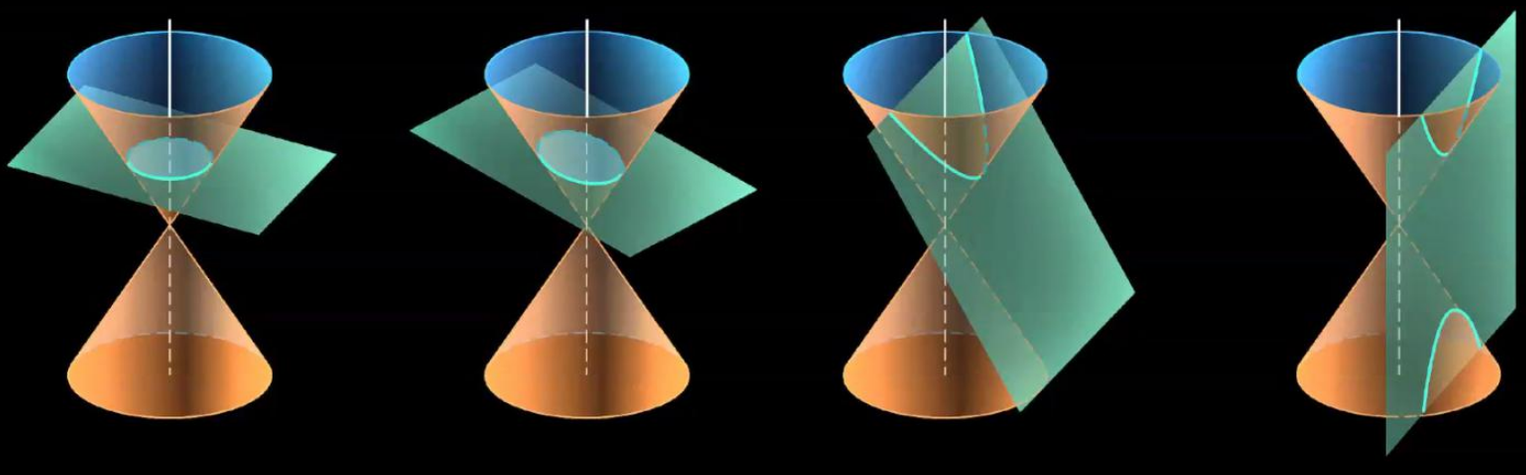
→ Every point on ellipse maintains constant ratio of distances from a point (Focus) and a line (directrix). This ratio is called eccentricity (e) $e = l_1/l_2$.

→ Sum of distances from any point on ellipse to two particular points is constant. These two points are called Foci, mathematically $S_1 + S_2 = 2a$.



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Conic Sections:

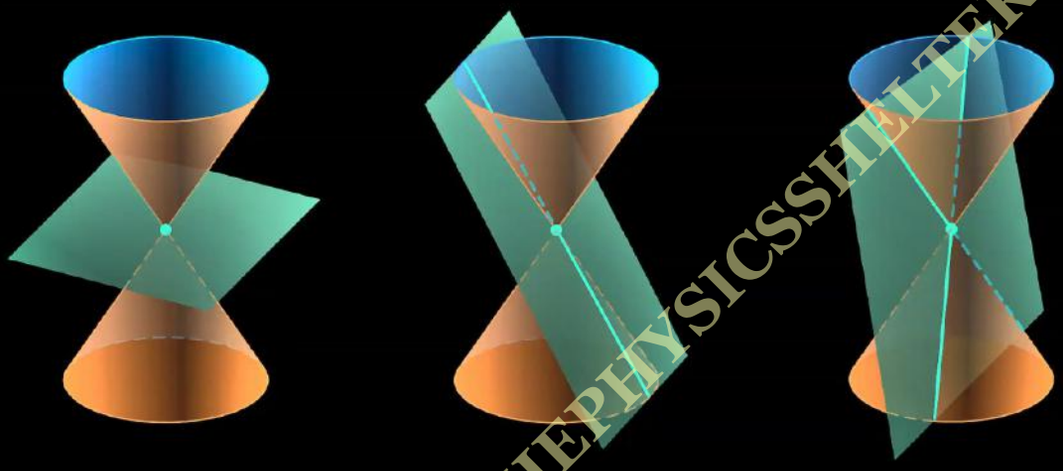


Circle: plane perpendicular to cone axis

Ellipse: plane oblique to cone axis

Parabola: plane parallel to side of cone

Hyperbola: plane parallel to cone axis



Point: plane through cone vertex only

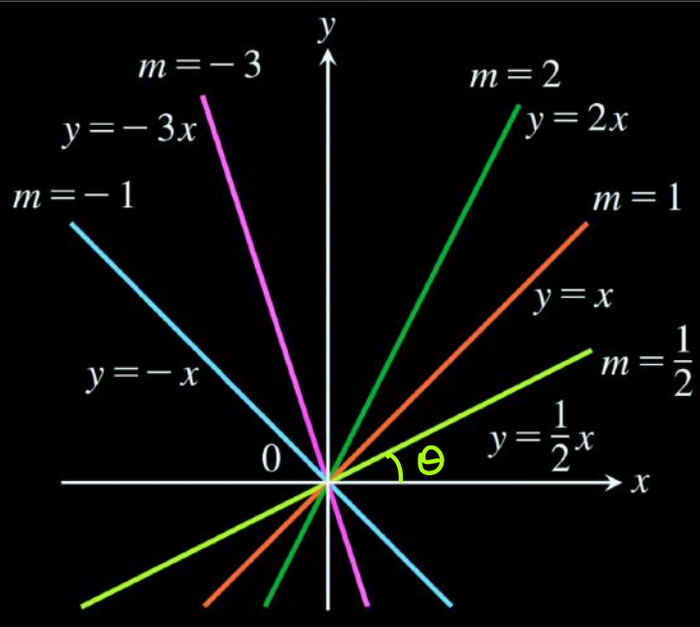
Single line: plane tangent to cone

Pair of intersecting lines

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Plots of straight lines:

A function of the form $y = mx + c$ represents a straight line. Here m is called the slope and c represents the intercept on y axis. Larger the magnitude of m, steeper the line. The adjacent figure shows several straight liners with $c = 0$ i.e passing through origin. Physically m represents the tan of the angle that the upper branch of line makes with positive x-axis. For example for the line in yellowish green has $\tan\theta = m = 1/2$.



Summary Of Important Formulae:

Quadratic Equation

Roots of $ax^2 + bx + c = 0$ are $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Sum of roots $x_1 + x_2 = -\dfrac{b}{a}$;

Product of roots $x_1 x_2 = \dfrac{c}{a}$

Binomial Theorem

$(1+x)^n = 1 + nx + \dfrac{n(n-1)}{2}x^2 + \dfrac{n(n-1)(n-2)}{6}x^3 + \ldots.$

$(1-x)^n = 1 - nx + \dfrac{n(n-1)}{2}x^2 - \dfrac{n(n-1)(n-2)}{6}x^3 + \ldots..$

If $x \ll 1$ then $(1+x)^n \approx 1 + nx$ & $(1-x)^n \approx 1 - nx$

Logarithm

$\log mn = \log m + \log n$ $\quad \log\dfrac{m}{n} = \log m - \log n$

$\log m^n = n \log m$ $\quad \log_e m = 2.303 \log_{10} m$

$\log_{10} 2 = 0.3010$ $\quad \log_{10} 3 = 0.4771$

Componendo and dividendo theorem

If $\dfrac{p}{q} = \dfrac{a}{b}$ then $\dfrac{p+q}{p-q} = \dfrac{a+b}{a-b}$

Arithmetic progression-AP

a, a+d, a+2d, a+3d, ...a+(n − 1)d
here d = common difference

Sum of n terms $S_n = \dfrac{n}{2}[2a+(n-1)d]$

Note: (i) $1+2+3+4+5....+ n = \dfrac{n(n+1)}{2}$

(ii) $1^2+2^2+3^2+...+ n^2 = \dfrac{n(n+1)(2n+1)}{6}$

Geometrical progression-GP

a, ar, ar^2, ar^3 ..... here, r = common ratio

Sum of n terms $S_n = \dfrac{a(1-r^n)}{1-r}$

Sum of ∞ terms $S_\infty = \dfrac{a}{1-r}$ [where $|r| < 1$]

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IMPORTANT TRIGONOMETRIC FORMULAE :

(i) $\sin(A + B) = \sin A \cos B + \cos A \sin B.$

(ii) $\sin(A - B) = \sin A \cos B - \cos A \sin B.$

(iii) $\cos(A + B) = \cos A \cos B - \sin A \sin B$

(iv) $\cos(A - B) = \cos A \cos B + \sin A \sin B$

(v) $\tan(A + B) = \dfrac{\tan A + \tan B}{1 - \tan A \tan B}$

(vi) $\tan(A - B) = \dfrac{\tan A - \tan B}{1 + \tan A \tan B}$

(vii) $\cot(A + B) = \dfrac{\cot B \cot A - 1}{\cot B + \cot A}$

(viii) $\cot(A - B) = \dfrac{\cot B \cot A + 1}{\cot B - \cot A}$

(ix) $2 \sin A \cos B = \sin(A+ B) + \sin(A - B).$

(x) $2 \cos A \sin B = \sin(A + B) - \sin(A - B).$

(xi) $2 \cos A \cos B = \cos(A + B) + \cos(A - B)$

(xii) $2 \sin A \sin B = \cos(A - B) - \cos(A + B)$

(xiii) $\sin C + \sin D = 2 \sin\left(\dfrac{C+D}{2}\right)\cos\left(\dfrac{C-D}{2}\right)$

(xiv) $\sin C - \sin D = 2 \cos\left(\dfrac{C+D}{2}\right)\sin\left(\dfrac{C-D}{2}\right)$

(xv) $\cos C + \cos D = 2 \cos\left(\dfrac{C+D}{2}\right)\cos\left(\dfrac{C-D}{2}\right)$

(xvi) $\cos C - \cos D = 2 \sin\left(\dfrac{C+D}{2}\right)\sin\left(\dfrac{D-C}{2}\right)$

(xvii) $\sin 2\theta = 2 \sin\theta \cos\theta = \dfrac{2\tan\theta}{1+\tan^2\theta}$

(xviii) $\cos 2\theta = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta = \dfrac{1-\tan^2\theta}{1+\tan^2\theta}$

(xix) $1 + \cos 2\theta = 2\cos^2\theta$ or $|\cos\theta| = \sqrt{\dfrac{1+\cos 2\theta}{2}}$

(xx) $1 - \cos 2\theta = 2\sin^2\theta$ or $|\sin\theta| = \sqrt{\dfrac{1-\cos 2\theta}{2}}$

(xxi) $\tan\theta = \dfrac{1-\cos 2\theta}{\sin 2\theta} = \dfrac{\sin 2\theta}{1+\cos 2\theta}$

(xxii) $\tan 2\theta = \dfrac{2\tan\theta}{1-\tan^2\theta}$

(xxiii) $\sin 3\theta = 3\sin\theta - 4\sin^3\theta.$

(xxiv) $\cos 3\theta = 4\cos^3\theta - 3\cos\theta.$

(xxv) $\tan 3\theta = \dfrac{3\tan\theta - \tan^3\theta}{1-3\tan^2\theta}$

(xxvi) $\sin^2 A - \sin^2 B = \sin(A+B).\sin(A-B) = \cos^2 B - \cos^2 A$

(xxvii) $\cos^2 A - \sin^2 B = \cos(A+B).\cos(A-B).$

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